

Human gut microbiota and its relationship to health and disease.

Probiotics are live microorganisms that confer a health benefit on the host when administered in appropriate amounts. Over 700 randomized, controlled, human studies have been conducted with probiotics thus far, with the results providing strong support for the use of probiotics in the clinical prevention or treatment of gastrointestinal tract disorders and metabolic syndrome. The present review is based on webinar presentations that were developed by the American Gastroenterological Association (AGA) in partnership with the International Scientific Association for Probiotics and Prebiotics (ISAPP) and the North American branch of the International Life Sciences Institute (ILSI North America). The presentations provided gastroenterologists and researchers with fundamental and current scientific information on the influence of gut microbiota on human health and disease, as well as clinical intervention strategies and practical guidelines for the use of probiotics and prebiotics. © 2011 International Life Sciences Institute.